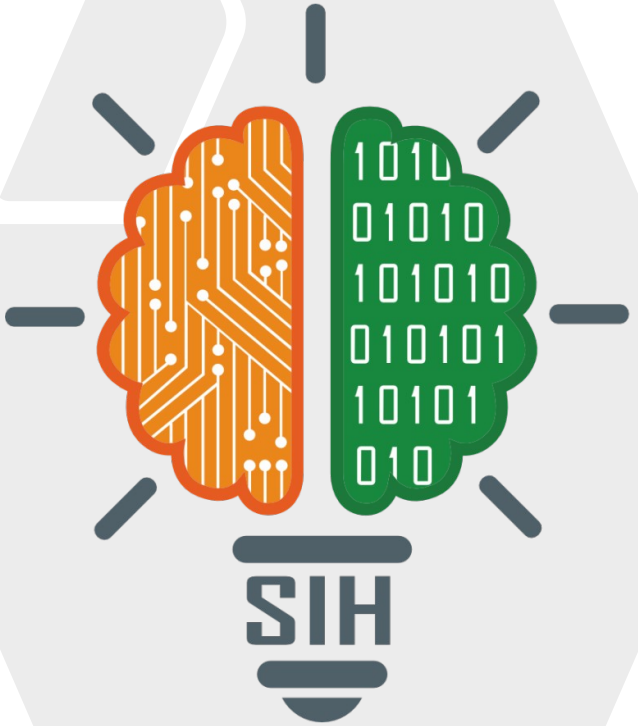
SMART INDIA HACKATHON 2026

SCOURGE

Ask the bank, not the paper.



- **Problem Statement ID –** SIH26265
- **Problem Statement Title –** Streamline of Security Deposit / FDR Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 199322
- **Team Name (Registered on portal) –** Arnavax#

SCOURGE

SCOURGE — ASK THE BANK, NOT THE PAPER

SMART INDIA HACKATHON 2026

Proposed Solution (Describe your Idea/Solution/Prototype)

The bank's own website proves the FDR: no bank API, no agreement, no paper. AICTE sees only that record.

Detailed explanation of the proposed solution

AICTE sends a **single-use challenge** for one FDR. The institution opens its own net banking; the connection key is **split with AICTE's verifier** (MPC-TLS), so nobody can fake the page. Login, OTP and PAN stay with the institution.

How it addresses the problem

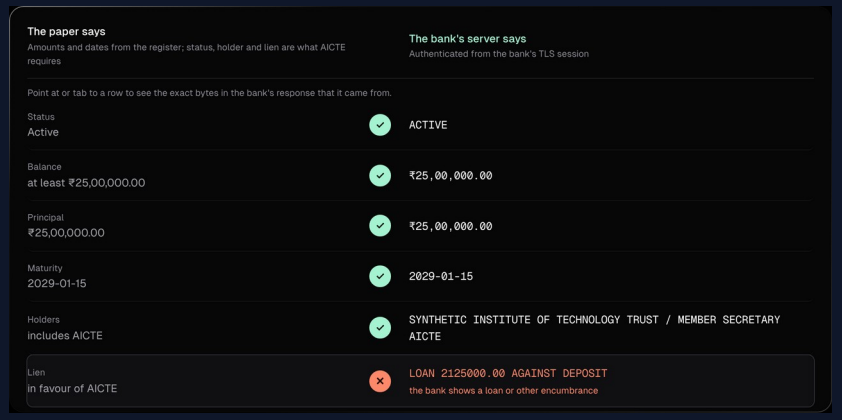
Paper is checked once and trusted for 10 years. Scourge **re-checks the bank any time and before every release**: a loan, early encashment, missing pledge (lien) or fake FDR shows up as a **Discrepancy**.

Innovation and uniqueness of the solution

Works with banks' existing net banking: no bank API or agreement needed, and API-ready when banks offer one. If a page cannot show a field: bank-signed e-advice, then branch confirmation. Fakes and replays are rejected.

LIVE PROTOTYPE · SYNTHETIC BANK

The bank's server contradicts the paper.



× HIDDEN LOAN CAUGHT

Every value traces to the bank's own bytes

The complete FDR lifecycle in one portal (all five stages run in the prototype)

1 FILE
Institution files online; renewals linked

→

2 PROVE
Bank-session proof on AICTE's challenge

→

3 MONITOR
T-90/60/30/7 alerts; adequacy; MT940/camt.053 reconcile

→

4 DECIDE
Verdict + evidence; hash-chained audit

→

5 PLEDGE RELEASE
Dept reviewers, 2 signers, holds; signed letter → bank ack

Why now: **₹201.42 crore of FDRs "do not appear in IDFC's records"** (CBI chargesheet, Oct 2026); AICTE joint FDRs found encashed or loaned against without consent.

SCOURGE

TECHNICAL APPROACH

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Technologies to be used

Proof layer
TLSNotary MPC-TLS · Rust
Selective disclosure · rustls

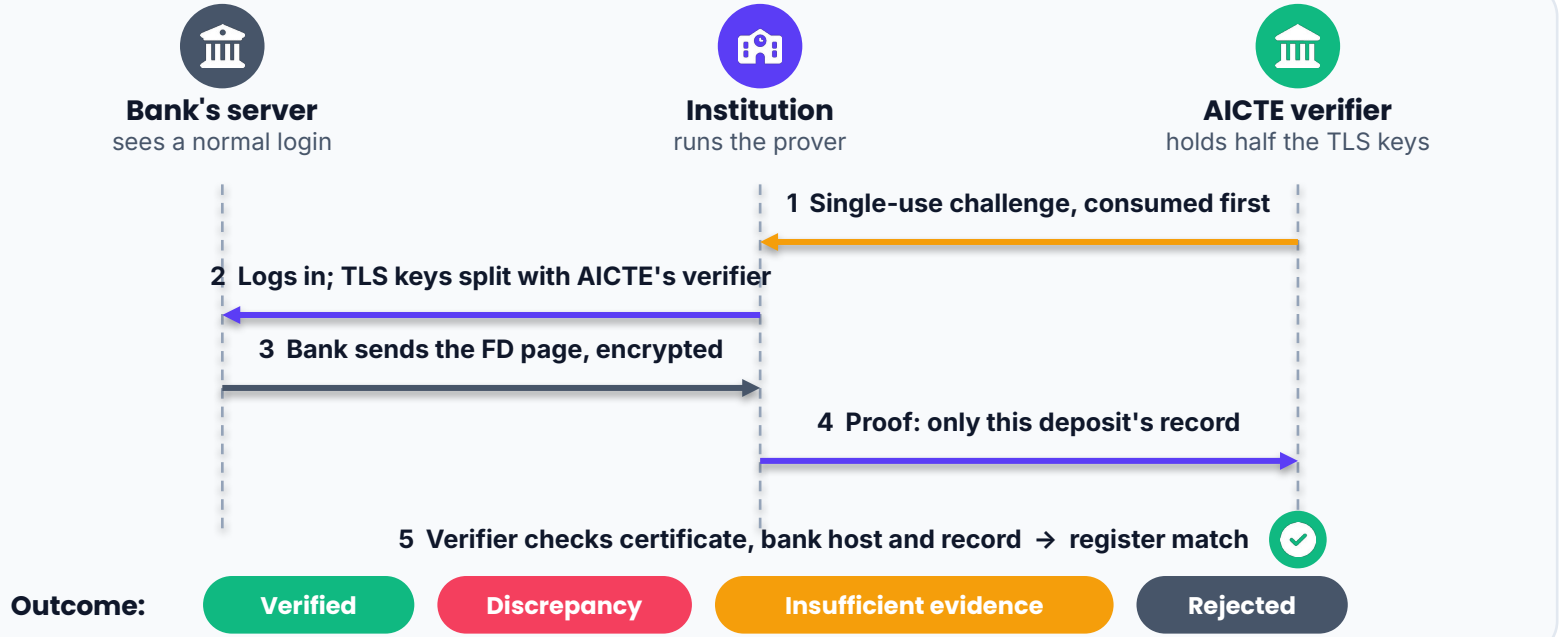
Backend
Python · FastAPI · SQLite
AES-GCM vault · hash-chained audit

Frontend
Server-rendered HTML · JS
Live 3-party proof view

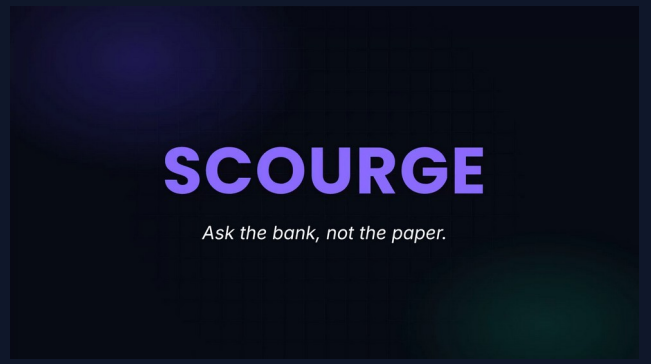
Ops & QA
Docker · nginx · GitHub CI
266 tests · 2FA · audits

Methodology and process for implementation (Flow Charts/Images/ working prototype)

In plain words: *an AICTE "notary" watches the bank page load, sees only the FD line, and nobody can change it.*



WATCH THE 3-MINUTE DEMO



Click to play here, or scan / click the QR for YouTube
Synthetic bank + live real-bank session

SCOURGE

FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.58 s
per bank proof
median of 60 runs · synthetic bank

Real bank
live split-key session
HTTP 200 in 1.0 s · first bank: SBI

266 / 266
automated tests pass
incl. rollback, forgery, backup-restore

₹0
software licences
60 MB/proof (measured) · FEASIBILITY.md

LIVE ON A REAL BANK
First bank tested: SBI · 30 Oct 2026

```
{"ok":true,"server":"api.onlineyono.sbi.bank.in","status_line":"HTTP/1.1 200 OK","secs":1.0}
```

Split-key session with a bank's production net-banking API; cert checked against public roots. Log: tools/realbank/RESULTS.md

Potential challenges and risks

Strategies for overcoming these challenges

Bank page can't be proven, or hides the lien	3-tier fallback: bank-signed e-advice, then branch; never a guess
Fake bank site or a replayed old proof	Rejected: bank certificate + single-use, 15-min challenge
Bank terms, consent and data protection	Own login, per-deposit consent, record-only disclosure; legal review
Bank encrypts data inside HTTPS, or ties it to the browser	Browser-extension prover; verifier gets the per-session app key

PILOT PLAN ✓ Prototype, 266 tests → ✓ Live real-bank session → FD pages at 3 large banks → 1 AICTE regional office → National rollout

SCOURGE

IMPACT AND BENEFITS

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Potential impact on the target audience

AICTE RB officers
A dated proof from the bank in seconds, not a letter in weeks

AICTE finance
The whole FDR book is re-checkable, before every release

Institutions
One short proof from their own net banking; Hindi or English

Students
29.6 lakh enrolled in 2025-26 (Lok Sabha reply); deposits checked all 10 years

Benefits of the solution (social, economic, environmental, etc.)

TODAY	WITH SCOURGE
Branch letter or uploaded screenshot	**The bank's own server, proven**
Checked once, trusted for 10 years	**Re-checked any time, before every release**
Weeks per confirmation	**0.58 s per proof (synthetic); 1.0 s live bank session**
Hidden loans found at release, or never	**Pledge checked at the bank; two-officer release**

TESTED ON 180 FRESH DEPOSITS
Random amounts, dates, formats · real proofs · synthetic bank

119 / 119	bad deposits the upload portal approved
0 / 119	bad deposits Scourge verified
0 / 61	honest deposits Scourge accused

Social
students' deposits stay intact

Economic
stops loss before release

Environmental
no paper letters or couriers

Governance
accounts, 2FA, vault, chained audit

SCOURGE

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Details / Links of the reference and research work

The problem is real

CBI: ₹201.42 cr of FDRs not in IDFC's records (ThePrint, 7 Oct 2026)
theprint.in › cbi-says-haryana-pollution-board...

CBI on AICTE: joint FDRs encashed or loaned without consent
deccanherald.com › greater-transparency-aicte...

Regulation

AICTE guidelines 2024-27: cl. 6.2(g) 10-year release; cl. 11.8
indiankanoon.org/doc/173919027

Account Aggregator users must be regulated FIUs (DFS)
financialservices.gov.in › account-aggregator...

Technology

TLSNotary: MPC-TLS proofs of web data, MIT / Apache-2.0
tlsnotary.org

NeSL e-BG for security deposits: new instruments only (WDRA)
wdra.gov.in › ebg_requirement-27-08-2025

PROJECT RESOURCES

Demo video (90 s): youtu.be/8HoplhFOGCg
Prototype: Scourge v1.0.0-rc1 · 266 / 266 tests · synthetic bank
Code + evidence: github.com/Lakki007/Arnavax-Scourge-SIH26265

Scourge vs existing approaches

	Upload & approve	Bank letter	Account Aggregator	NeSL e-BG	Scourge
Uses the bank's own data	✗	✓	✓	✓	✓
No bank API or agreement	✓	✓	✗	✗	✓
Institution cannot forge it	✗	~	✓	✓	✓
Covers existing FDR book	✓	✓	✓	✗	✓
Re-check in seconds	✗	✗	✓	✓	✓
Shown on a real bank today	✓	✓	✓	✓	Live bank session; FD next

~ a bank letter can still be forged. Live split-key session with a real bank done (SBI, 30 Oct 2026); FD-page proof is roadmap step 1.